Hyeongmuk LIM

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RESEARCH INTERESTS

Topological phases of quantum matter; quantum geometry of multiband electronic states; geometric and topological approaches to time-reversal-symmetric correlated systems. Broader interests include strongly correlated and many-body quantum physics.

EDUCATION

Integrated M.S./Ph.D. Program in Physics Expected August 2027
Seoul National University, Seoul, Republic of Korea

- ✓ *Advisor*: Prof. Bohm-Jung Yang
- ✓ *GPA*: 3.7/4.0
- ✓ *Related coursework*: Theoretical and computational condensed matter physics, quantum field theory, algebraic geometry, special topics in mathematical physics

Bachelor's Studies in Physics & Mathematics 2021
Seoul National University, Seoul, Republic of Korea

- ✓ *GPA*: 4.06/4.3
- ✓ *Related coursework*: Double major in Physics and Mathematics, graduate coursework in differential geometry and algebraic topology
- ✓ *Thesis (Physics)*: A construction of PT-symmetric 4-band insulator using Hopf map
- ✓ *Thesis (Mathematics)*: Survey on thermodynamic formalism and dynamical systems approach to statistical mechanics

PUBLICATIONS & PREPRINTS

- ✓ **Hyeongmuk Lim**, J. Jung, Y. Qian, B.-J. Yang (2026). Quaternion-Kähler geometry of time reversal symmetric crystals. arXiv:2608.03178 (preprint)
- ✓ S. Kim, Y. Choi, **Hyeongmuk Lim**, B.-J. Yang (2024). Unconventional topological phase transition of the Hopf insulator. arXiv:2410.04021 (preprint)
- ✓ J. Jung, **Hyeongmuk Lim**, B.-J. Yang (2024). Quantum geometry and Landau levels of quadratic band crossings. *Physical Review B* **109**, 035134. DOI: 10.1103/PhysRevB.109.035134
- ✓ **Hyeongmuk Lim**, S. Kim, B.-J. Yang (2023). Real Hopf insulator. *Physical Review B* **108**, 125101. DOI: 10.1103/PhysRevB.108.125101

PRESENTATIONS

Conference Presentations

- ✓ Classification of delicate and multigap topological bands protected by a combination of space inversion and additional discrete symmetries, *APS Global Physics Summit 2025* (contributed oral presentation, MAR-L21.00001), Anaheim, USA (19 March 2025). APS abstract
- ✓ Real Hopf insulator, Quantum Materials Symposium, Pyeongchang, Republic of Korea (21 February 2024)
- ✓ Real Hopf insulator, ExU-YITP Workshop on Condensed Matter Physics and Quantum Information, Kyoto, Japan (28 September 2023)

AWARDS & FELLOWSHIPS

- ✓ BK우수연구인재 Fellowship (2022), Seoul National University
- ✓ 4단계 BK21 장학금 (2022–2024), Seoul National University